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2024

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How to cite

KARASAN, Duygu. L'influence de la géométrie et de la hauteur du pilier titane-base sur la stabilité mécanique des couronnes uniques supportées par implant. Doctoral Thesis, 2024. doi: 10.13097/archive-ouverte/unige:180395

This publication URL: <https://archive-ouverte.unige.ch/unige:180395>

Publication DOI: [10.13097/archive-ouverte/unige:180395](https://doi.org/10.13097/archive-ouverte/unige:180395)

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**UNIVERSITÉ
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DE GENÈVE**

FACULTÉ DE MÉDECINE

Clinique universitaire de médecine dentaire

Département de réhabilitation oro-faciale

Division de prothèse fixe et biomatériaux

Mémoire préparé sous la direction de la professeure **Irena SAILER**

**L'influence de la géométrie et de la hauteur du pilier titane-base sur
la stabilité mécanique des
couronnes uniques supportées par implant**

Thèse

présentée à la Faculté de Médecine de
l'Université de Genève

pour obtenir le grade de Docteur en médecine dentaire
par

Duygu KARASAN

de
Ankara (Turquie)

Thèse n°

Genève

2023



**UNIVERSITÉ
DE GENÈVE**
FACULTÉ DE MÉDECINE

DOCTORAT EN MEDECINE DENTAIRE

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Intitulée :

L'influence de la géométrie et de la hauteur du pilier titane-base sur la stabilité mécanique des couronnes uniques supportées par implant

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REMERCIEMENTS

As one attains new milestones in their career, accomplishments may begin to feel ordinary, and these moments often give the feeling of checking another item off the list.

However, upon reaching certain milestones, I choose to pause and reflect, appreciating the journey and recalling how fortunate I am to have completed this work with the support of incredible individuals I am privileged to have in my life and collaborate with.

With that in mind, I would like to express my gratitude to the following:

Prof Irena Sailer, your continuous support and daily life lessons are invaluable. I am genuinely grateful for your guidance, both personally and professionally; it is never taken for granted.

Vincent Fehmer, thank you for your constant availability and wise counsel. Your advice has taught me the importance of remaining calm and composed in all situations.

Dr. Cristina Zarauz, you share your knowledge, clinical experience any day, anytime, without restraint and remind me every day of the importance of exploring the depths of knowledge, looking beyond the rushing and racing that academic life dictates. You've supported me through the most difficult professional and personal periods with such a stoicism; I often find myself thinking about what you would do in a given situation. I greatly appreciate having you by my side in my journey.

All my co-authors, thank you for your unwavering support throughout this investigation and for helping see this work through to completion.

Dr. Agnes Sahli, you have become an inspiration with your work ethics, discipline, enormous energy, honesty, trustworthiness and humility. Thanks for editing the french sections of this thesis.

My colleagues from the Division de Prothèse Fixe et Biomatériaux, your friendship makes the work struggles more bearable.

To my husband Mustafa, you are my rock, I am grateful for your unwavering support and love.

To my daughter Ayse, you literally give meaning to everything in life; nothing would be the same without you.

RÉSUMÉ EN FRANÇAISE

OBJECTIF

Cette étude vise à examiner l'influence de la macro et micro-géométrie des piliers en titane (ti-base) sur la stabilité mécanique des couronnes individuelles supportées par implant, vissées, réalisées en réseau céramique infiltrée par polymère (PICN).

MATÉRIAUX ET MÉTHODES

Douze spécimens par groupe ont été utilisés, comprenant six combinaisons différentes d'implants/piliers ti-base restaurées avec des couronnes individuelles en PICN supportées par implant: Nb-T (hauteur gingivale [HG] : 1,5 mm, hauteur prothétique [HP] : 4,3 mm), CC (HG : 0,8 mm, HP : 4,3 mm), CC-P (HG : 0,8 mm, HP : 7 mm), Nb-V (HG : 1,5 mm, HP : 6 mm), St (HG : 1,5 mm, HP : 5,5 mm) et Th (HG : 0,5 mm, HP : 9 mm). Les spécimens ont subi un vieillissement thermomécanique, et ceux qui ont survécu ont ensuite été soumis à une charge statique jusqu'à la défaillance. Les données ont été analysées à l'aide d'un test ANOVA, suivi du test post hoc de Tukey ($\alpha = 0,05$).

RÉSULTATS

Tous les spécimens ont survécu au vieillissement thermomécanique sans complications, à savoir des fissures visibles, un désencollage ou un desserrage de la vis. Le groupe Th a montré les valeurs de résistance les plus élevées parmi tous les groupes, avec des différences significatives par rapport à Nb-T ($p < 0,05$), CC ($p < 0,001$) et St ($p < 0,001$). De plus, le groupe CC-P a présenté des résultats de résistance à la fracture significativement supérieurs par rapport à CC ($p < 0,05$) et St ($p < 0,05$).

CONCLUSION

Le choix du ti-base, en particulier la hauteur prothétique, a eu une influence significative sur la résistance à la fracture des couronnes individuelles en PICN. Cependant, la hauteur ou les caractéristiques géométriques du ti-base n'ont pas exercé une influence significative sur le comportement mécanique de l'assemblage couronne/ti-base sous charge thermomécanique, car tous les spécimens ont résisté au vieillissement sans complication ni défaillance.

ABSTRACT IN ENGLISH

AIM

This study aimed to investigate the influence of titanium base (ti-base) abutment macro and micro geometry on the mechanical stability of polymer-infiltrated ceramic network (PICN) screw-retained implant-supported single crowns (iSCs).

MATERIALS AND METHODS

Twelve specimens per group were used, comprising six different implant/ti-base abutment combinations restored with PICN iSCs: Nb-T (gingival height [GH]: 1.5 mm, prosthetic height [PH]: 4.3 mm), CC (GH: 0.8 mm, PH: 4.3 mm), CC-P (GH: 0.8 mm, PH: 7 mm), Nb-V (GH: 1.5 mm, PH: 6 mm), St (GH: 1.5 mm, PH: 5.5 mm), and Th (GH: 0.5 mm, PH: 9 mm). The specimens underwent thermo-mechanical aging, and those that survived were subsequently subjected to static loading until failure. The data was analysed using a one-way ANOVA test followed by Tukey post hoc test. ($\alpha = .05$)

RESULTS

All specimens survived thermo-mechanical aging without complications namely visible cracks, debonding, or screw loosening. Th group demonstrated the highest strength values among all the groups, with significant differences compared to Nb-T ($p < .05$), CC ($p < .001$), and St ($p < .001$). Additionally, CC-P group exhibited significantly superior fracture strength results compared to CC ($p < .05$) and St ($p < .05$).

CONCLUSION

The choice of ti-base, particularly prosthetic height had a significant influence on fracture resistance of PICN iSCs. Nevertheless, the height or geometrical features of the ti-base did not exhibit a significant influence on the mechanical behaviour of the iSC/ti-base assembly under thermomechanical loading, as all specimens withstood the aging without complication or failure.

1. INTRODUCTION

L'évolution de l'implantologie dentaire a connu un changement de paradigme significatif avec le développement de matériaux de restauration monolithiques et l'avènement des piliers en base de titane (Ti-base). Les piliers en base de titane se sont révélés être des éléments cruciaux dans les restaurations implantaires, offrant aux cliniciens une gamme étendue d'options prothétiques pour améliorer la qualité des résultats. Bien que la littérature existante mette en avant les avantages des piliers en base de titane, il est de plus en plus nécessaire de comprendre plus précisément l'interaction complexe entre leur stabilité mécanique et leurs caractéristiques géométriques.

1.1. Contexte

1.1.1. Couronnes Unitaires sur Implants (iSCs)

La pose d'implants suivie de la restauration avec une couronne unitaire supportée par implant (iSC) est un traitement largement accepté pour remplacer une seule dent manquante (Wittneben, Joda et al. 2017, Rabel, Spies et al. 2018), affichant des taux de survie dépassant 90 % sur une période de 5 ans, comme le montrent des revues systématiques antérieures (Jung, Zembic et al. 2012, Larsson and Wennerberg 2014, Abou-Ayash, Strasding et al. 2017, Pjetursson, Valente et al. 2018, Rabel, Spies et al. 2018). Ces résultats cliniques positifs ont conduit à une utilisation généralisée des couronnes supportées par implant en tant qu'alternative viable aux prothèses dentaires fixes multi-éléments supportées par les dents naturelles, s'alignant sur la tendance contemporaine de préserver les dents naturelles grâce à des procédures peu invasives. Un facteur pouvant influencer les résultats des iSCs est le matériau à partir duquel les restaurations sont fabriquées. Alors qu'auparavant, les restaurations en métal-céramique étaient principalement utilisées, de nos jours, une myriade de matériaux tout-céramique ou hybrides-céramique sont disponibles.

1.1.2. Sélection des Matériaux pour iSC

Les couronnes implantaires monolithiques en céramique pour protocole de guérison implantaire ou couronnes implantaires monolithiques (iSC) en céramique-frittée sur métal sont traditionnellement considérées comme la norme, et les restaurations en céramique-frittée sur métal ont dominé les applications cliniques pendant

longtemps, avec des taux de survie des iSC sur 5 ans élevés de 95,8 % (Jung, Zembic et al. 2012). Néanmoins, les iSC entièrement en céramique gagnent en popularité en raison de leur attrait esthétique et de leur aspect naturel. Les premiers matériaux entièrement en céramique introduits (par exemple, la céramique feldspathique, la leucite pressée et les céramiques vitrifiées renforcées à l'alumine) ne pouvaient pas rivaliser avec la stabilité mécanique des iSC en céramique-frittée sur métal dans les zones soumises à des charges élevées dans l'arcade dentaire. Leur indication était donc limitée aux zones esthétiquement exigeantes au début (Pjetursson, Sailer et al. 2021). Les améliorations dans les technologies dentaires numériques ont ouvert des voies alternatives aux processus de fabrication conventionnels, facilitant l'utilisation d'une gamme plus large de matériaux céramiques, notamment la céramique oxyde de haute résistance (zircone [ZrO₂] et alumine [Al₂O₃]), les céramiques vitrifiées renforcées (disilicate de lithium [Li₂Si₂O₇] et silicate de lithium renforcé de zircone [Li₂SiO₃]), et des matériaux hybrides tels que les céramiques à matrice de résine (RMC) (composites à base de résine) et les matériaux en réseau céramique infiltrée de polymère [PICN] (Spitznagel, Boldt et al. 2018). À mesure que les flux de travail numériques deviennent plus efficaces (Mühlemann, Kraus et al. 2018), les indications pour les restaurations entièrement en céramique sont de plus en plus élargies. Les infrastructures en zircone de haute résistance, revêtues de céramique à des fins esthétiques (c'est-à-dire zircone avec stratification feldspathique), sont une alternative en céramique bien documentée aux céramiques sur métal pour les restaurations implantaires aujourd'hui (Pjetursson, Valente et al. 2018, Rabel, Spies et al. 2018). La revue systématique de Pjetursson et al. (2018), comparant les iSC en zircone avec stratification et en céramique métallique supportée par implant, a rapporté des taux de survie à 5 ans de 97,6 % (intervalle de confiance à 95 % (IC) : 94,3–99,0) et de 98,3 % (IC à 95 % : 96,8–99,1), respectivement, pour les deux types de iSC. Les taux de complications biologiques et techniques des iSC en zircone avec stratification et en céramique métallique étaient similaires. Cependant, la fracture de la couche de céramique feldspathique était le problème technique prédominant pour les deux types de restaurations stratifiées (Pjetursson, Valente et al. 2018). Une autre revue systématique a comparé les résultats des iSC en céramique oxyde et en céramique vitrifiée supportées par implant (Rabel, Spies et al. 2018). Les auteurs ont rapporté de bons taux de survie estimés à 5 ans pour les iSC en céramique supportées par implant (93 % (IC à 95 % : 86,6–96,4 %)), mais encore une fois, des taux élevés d'éclats de céramique de stratification atteignant 9,0 % (IC à 95 % : 5,4–14,8 %) sur une période de 5 ans (Rabel, Spies et al. 2018). Afin de surmonter les problèmes techniques rencontrés avec tous les types de restaurations

stratifiées, plus récemment, un type de restaurations monolithiques a été présenté (Caramês, Marques et al. 2019, Cheng, Chien et al. 2019, Rammelsberg, Lorenzo Bermejo et al. 2020). Ces restaurations de type monolithique peuvent offrir deux avantages principaux. Premièrement, l'augmentation de l'efficacité des procédures de laboratoire en permettant une fabrication plus rapide des restaurations (Joda and Brägger 2014, Joda and Brägger 2015, Joda and Brägger 2016). Deuxièmement, l'amélioration des résultats cliniques en réduisant le nombre de fractures céramiques, l'un des problèmes prédominant observé avec les restaurations stratifiées (Pjetursson, Sailer et al. 2021). Cependant, il convient de garder à l'esprit que l'importance clinique de ces éclats peut varier en fonction de l'emplacement et des caractéristiques des fractures céramiques. Les réseaux céramiques infiltrés de polymère (PICN) sont classés comme des composites à phases intercalées (IPC), présentant une géométrie tridimensionnelle interconnectée. L'introduction de matériaux intercalaires en céramique en dentisterie remonte à plus de deux décennies avec la création du système In-Ceram (Vita Zahnfabrik, Bad Saeckingen, Allemagne). Les matériaux In-Ceram utilisent une structure céramique poreuse avec une phase intercalaire en verre, comme l'a élucidé Horvitz et al. (Horvitz, Gotman et al. 2002), mettant en évidence les avantages de la matrice céramique (Al_2O_3) tels que la résistance à l'usure améliorée et la résistance à haute température. L'inclusion de la phase métallique augmente en outre la ténacité par le biais d'un pontage des fissures. Les phases interconnectées au sein des IPC, y compris les PICN, limitent la propagation des fissures en déviant les fissures interfaciales. La phase présentant une plus grande contrainte à la rupture contribue à la résistance à la fracture en reliant les fissures introduites à l'autre phase (Horvitz, Gotman et al. 2002, Nalla, Kinney et al. 2003, Coldea, Swain et al. 2013). Contrairement à In-Ceram, où la phase de verre cassante est présente, les PICN la remplacent par un polymère, formant deux réseaux intercalés continus. Les matériaux PICN CAD/CAM visent à résoudre les problèmes de fragilité, de rigidité et de dureté associés aux céramiques tout en améliorant la flexibilité, la ténacité à la rupture et la possibilité d'usinage. L'objectif est d'imiter le comportement mécanique des dents naturelles, avec des aspirations supplémentaires pour des caractéristiques d'usure semblables à celles des composites et de l'émail (Menini, Conserva et al. 2013).

1.1.3. Concept de la base en titane (Ti-base)

Les bases titane compatibles avec la CFAO sont devenues essentielles en implantologie, servant de plateforme

pour l'usinage précis de superstructures personnalisées. Une couronne sur implant supportée par une Ti-base se compose généralement d'un pilier en titane solidement connecté à l'implant, servant de fondation pour la placement ultérieur et le scellement d'une couronne personnalisée (E, Zahoui et al. 2021).

Les Ti-bases, disponibles dans diverses hauteurs et contours d'attachement rétenteurs transmuqueux, résident dans une bibliothèque numérique avec des fichiers STL ouverts contenant les géométries nécessaires. Une fois qu'une Ti-base est choisie et que la superstructure est conçue à l'aide d'un logiciel de CFAO, le fichier STL résultant est envoyé pour l'usinage. La surface de la restauration en lien avec la Ti-base nécessite généralement un ajustement minimal, ce qui assure une adaptation intime avant le scellement. Par la suite, la restauration peut être finalisée avant le scellement adhésif sur la Ti-base en dehors de la bouche (Chantler, Evans et al. 2023).

Tout en offrant la supériorité de la connexion implant/pilier préfabriquée, le concept de la Ti-base offre des avantages, notamment une esthétique améliorée, la capacité de reproduire un profil d'émergence sur mesure pour rétablir le contour anatomique des tissus mous, la réversibilité, des performances biomécaniques améliorées, l'absence de matériau céramique au niveau de la connexion implant-pilier, et une procédure de scellement en laboratoire contrôlée (Chantler, Evans et al. 2023).

1.1.4. Les facteurs influençant la stabilité mécanique des iSC supportées par des piliers en Ti-base

La stabilité mécanique de l'assemblage couronne/base en titane (Ti-base) n'est pas exempte de défis. Des facteurs ont été identifiés qui pourraient et/ou influencent la stabilité mécanique de l'assemblage Ti-base/iSC. La force de l'interface entre ces composants est un déterminant critique, avec des aspects de rétention et d'adhésion jouant des rôles clés. Des recherches approfondies menées par Rödiger et al. (2018), Ioannidis et al. (2020) et Zahoui et al. (2020) ont souligné les dynamiques complexes impliquées dans l'obtention d'une stabilité mécanique optimale (Rödiger, Kloß et al. 2018, Ioannidis, Gil et al. 2020, Zahoui, Bergamo et al. 2020).

Des études antérieures, dont celles de Pitta, Burkhardt, et al. (2021), Burkhardt et al. (2022), et Lopes et al. (2019), se sont penchées sur l'influence des traitements de surface et des protocoles adhésifs sur la stabilité mécanique et les forces de détachement des couronnes unitaires supportées par implant (iSC) (Lopes, Machado

et al. 2019, Pitta, Burkhardt et al. 2021, Burkhardt, Sailer et al. 2023). Le consensus de ces études souligne la nécessité d'un traitement de surface, en particulier par sablage, pour augmenter les forces de liaison et assurer la stabilité mécanique des iSC sur les bases en Ti. Burkhardt et al. 2022 ont rapporté qu'aucun des systèmes adhésifs ne fonctionnait généralement mieux que les autres sur tous les matériaux, ce qui signifie que pour chaque matériau de restauration, un système adhésif différent peut être recommandé. Bien que les primers des trois systèmes adhésifs testés contiennent du silane, qui s'est avéré bénéfique pour l'adhérence aux céramiques silicatées (Frankenberger, Hartmann et al. 2015), et du 10-méthacryloyloxydecyl-dihydrogen-phosphate (10-MDP) essentiel pour une liaison durable à la zircone et au titane (Kern and Wegner 1998, Nagaoka, Yoshihara et al. 2017), leur performance a divergé de manière significative. Par conséquent, les nouveaux matériaux CAD/CAM montrent des performances d'adhérence favorables avec différents systèmes adhésifs, cependant, pour chaque matériau de restauration, un système adhésif spécifique doit être recommandé, car aucun des systèmes n'a fonctionné de manière équivalente avec chacun des matériaux différents (Burkhardt, Sailer et al. 2023).

Bien que beaucoup d'attention ait été accordée aux traitements de surface et aux protocoles adhésifs, l'impact sur la stabilité de l'adhésion entre les iSC et les bases en Ti des facteurs de macro et micro-géométrie, incluant la hauteur prothétique du pilier, la surface adhésive de la Ti-base, les caractéristiques anti-rotatives, le diamètre de la partie prothétique et les caractéristiques de surface des bases en Ti, reste relativement inexploré.

Les tests de déscollement in vitro réalisés par Rödiger et al. (2018), Silva et al. (2018) et Zahoui et al. (2020) constituent la base de notre compréhension actuelle (Rödiger, Kloth et al. 2018, Silva, Soares et al. 2018, Zahoui, Bergamo et al. 2020). Cependant, l'impact plus important de la macro et micro-géométrie du pilier sur la stabilité mécanique globale de l'assemblage couronne-pilier est un domaine qui nécessite des recherches plus approfondies.

1.2. Objectif de l'étude et hypothèses nulles

Cette étude in vitro visait à évaluer l'influence combinée des facteurs de macro et microgéométrie des piliers en titane (ti-base) en testant six combinaisons différentes de piliers en ti-base et d'implants sur la stabilité mécanique de couronnes unitaires vissées réalisées en céramique infiltrée de polymère (PICN). En

conséquence, l'hypothèse nulle de cette étude est que « la géométrie du pilier n'aurait aucun effet sur la stabilité mécanique globale des couronnes unitaires vissées supportées par implant avec des piliers en ti-base ».

2. DISCUSSION

Les résultats de l'étude actuelle indiquent un impact notable de la hauteur prothétique de la base en titane (Ti-base) sur la résistance à la fracture des couronnes supportées par implant (iSC). Des facteurs tels que le sablage préalable de la surface de la base en titane et la présence de caractéristiques macro-géométriques anti-rotationnelles n'ont pas montré d'effet significatif sur la résistance à la fracture. Il est important de noter qu'aucune complication liée à la Ti-base, à la iSC ou à l'implant n'a été observée lors du test de vieillissement thermomécanique, conduisant à un taux de survie de 100 % et rejetant partiellement l'hypothèse nulle.

Tous les spécimens ont supporté le processus de vieillissement thermomécanique sans aucun signe visible de défaillance comme des fissures, un décollement de la couronne, un desserrage de la vis ou un micro-mouvement lié au déscellement. La charge appliquée de 98N pendant les tests a dépassé la force de mastication maximale anticipée à 70N dans la zone des molaires, même en l'absence d'activité parafunctionnelle. Des résultats comparables ont été observés dans une étude menée par Yazigi et al. (2020) examinant différents matériaux iSC dans des conditions de thermocyclage similaires, démontrant la robustesse des spécimens testés (Yazigi, Kern et al. 2020).

2.1. Géométrie de la ti-base

Notre étude a révélé que la hauteur prothétique de la Ti-base est un facteur critique influençant la résistance à la fracture de l'assemblage couronne-base en titane. Les piliers plus longs ont présenté des forces de résistance à la fracture significativement plus élevées que leurs homologues plus courts.

Cano-Batalla et al. ont rapporté des résultats qui correspondent aux observations de la présente étude (Cano-Batalla, Soliva-Garriga et al. 2012). La mise en évidence dans les deux études que la hauteur prothétique de la Ti-base émerge comme un facteur critique affectant la résistance à la fracture est d'un intérêt particulier. Les Ti-bases plus longues ont systématiquement montré des forces de résistance à la fracture plus élevées, une tendance conforme aux résultats de Sieper et al. (2017). Malgré des valeurs moyennes de résistance à la fracture

plus faibles pour les PICN par rapport au disilicate de lithium, les valeurs observées dans les deux études dépassent les forces masticatoires physiologiques, soulignant la fiabilité des spécimens testés (Cano-Batalla, Soliva-Garriga et al. 2012, Sieper, Wille et al. 2017, Yazigi, Kern et al. 2020).

Notre étude, tout comme les investigations précédentes, attribue des performances supérieures après le thermocyclage à un protocole de collage spécifique. Les traitements de surface et les protocoles de collage, identifiés comme des contributeurs cruciaux à la stabilité mécanique, ont été essentiels pour minimiser le déscollement dans les deux études. L'adoption d'un protocole adhésif (Clearfil Ceramic Primer Plus et Panavia V5) dans notre étude est conforme aux stratégies de collage efficaces identifiées dans la recherche de Burkhardt et al. (2022) (Burkhardt, Sailer et al. 2023).

De plus, l'impact accru de la hauteur prothétique sur la résistance à la fracture, comme démontré dans les deux études, souligne l'importance de ce paramètre par rapport à d'autres propriétés examinées telles que le sablage de surface ou les caractéristiques macro-géométriques anti-rotatives. La concordance des résultats et l'absence de complications lors du chargement thermomécanique soulignent le rôle critique de la hauteur prothétique dans les assemblages Ti-base/iSC.

2.2. Influence du protocole adhésif

La performance supérieure observée dans notre étude après le test de thermocyclage peut être attribuée au protocole de scellement spécifique utilisé. Conformément aux recherches antérieures, les traitements de surface et les protocoles de scellement ont été identifiés comme des contributeurs cruciaux à la stabilité mécanique de l'assemblage iSC/Ti-base (Pitta, Burkhardt et al. 2021, Burkhardt, Sailer et al. 2023). Notre étude a adopté un système de ciment résineux (Clearfil Ceramic Primer Plus et Panavia V5), qui s'est avéré efficace pour minimiser le déscollement, soulignant davantage l'importance de protocoles de scellement adaptés pour des performances optimales (Burkhardt, Sailer et al. 2023).

De plus, les résultats des forces de traction rapportés par Burkhardt et al. ont révélé des valeurs accrues (plus de 500 N) pour les trois matériaux de restauration différents en fonction du système de collage utilisé. Fait intéressant, aucun des systèmes de collage n'a démontré une supériorité constante sur tous les matériaux, suggérant que pour chaque matériau de restauration, un système de collage différent peut être recommandé.

Malgré la présence de silane et de 10-méthacryloyloxydecyl-dihydrogène-phosphate (10-MDP) dans les primers, connus pour leurs propriétés bénéfiques d'adhérence aux céramiques silicatées et au titane, respectivement, la performance a divergé significativement entre les systèmes de scellement testés. Cette divergence pourrait être attribuée aux variations dans les compositions, concentrations et formulations chimiques des primers dans chaque système de scellement, mettant en évidence la nature nuancée et spécifique au matériau d'un collage efficace dans les applications dentaires.

2.3. Matériau prothétique

Dans la présente étude, les charges de fractures observées dans chaque groupe de spécimens ont dépassé 2000 N, dépassant significativement la plage typique de forces de mastication de 346 à 548 N (Karasan, Canay et al. 2022). Cette force notable peut être attribuée, en partie, aux propriétés mécaniques du matériau en réseau céramique infiltré de polymère (PICN). Des recherches suggèrent que les matériaux PICN présentent une ténacité accrue et une usinabilité supérieure par rapport aux céramiques, en raison de leur réduction de la fragilité, de la rigidité et de la dureté, associée à une flexibilité accrue (Coldea, Swain et al. 2013). De plus, l'application de ce matériau céramique hybride aux caractéristiques modérément élastiques dans les couronnes supportées par implant (iSCs) a démontré une capacité significative à absorber ou diminuer d'environ 70 % des forces de mastication transmises à l'os péri-implantaire simulé (Menini, Conserva et al. 2013). Cette amélioration substantielle contraste avec la rigidité généralement associée à des matériaux tels que la zircone, la céramique vitreuse et l'or.

Burkhardt et al. (2022) ont étudié trois matériaux de restauration différents avec trois systèmes adhésifs, notant que les couronnes PICN étaient généralement plus résistantes aux altérations marginales et moins susceptibles de présenter des micro-mouvements par rapport à la disilicate de lithium et à la zircone en tant que matériaux de restauration utilisés pour l'assemblage ti-base/SC (Burkhardt, Sailer et al. 2023). Cette amélioration de l'intégrité marginale et des performances de l'interface de liaison pourrait s'expliquer par le fait que le PICN a un module d'élasticité plus élevé (Homaei, Farhangdoost et al. 2016), lui permettant d'absorber plus d'énergie et entraînant un meilleur amortissement des forces occlusales par rapport aux céramiques d'oxyde ou de silicate (Magne, Silva et al. 2013). Ces observations sont en accord avec les résultats de notre étude, contribuant à une

compréhension complète des caractéristiques mécaniques du PICN et de ses implications pour la stabilité et les performances des assemblages ti-base/iSC.

2.4. Conditions de test

L'étude a soigneusement mis en œuvre des conditions d'essai standardisées, soumettant les spécimens à un régime simultané de thermocyclages (5°–55°C, temps de séjour 120 s) et à une simulation de mastication impliquant 1 200 000 cycles à 98 N et 1,67 Hz (CS-4.8, SD Mechatronik). Les spécimens ont été chargés depuis la surface occlusale, à 2 mm lingual du canal d'accès à la vis, en maintenant le parallélisme avec l'axe de l'implant, et avec un mouvement vertical défini de 2 mm. L'utilisation de billes en céramique (Ø 6 mm, bille de stéatite, Hoechst CeramTec), reproduisant la dureté Vickers de l'émail, en tant qu'antagonistes, a amélioré la spécificité de la simulation. Ensuite, une charge statique jusqu'à la défaillance, exécutée avec une machine d'essai universelle de la série Shimadzu AGS-X.

Une force importante de l'étude réside dans la définition méticuleuse des critères de défaillance. Que ce soit en discernant une fracture visible du pilier ou de la couronne ou en détectant une diminution de 20 % de la charge maximale pendant l'essai de charge jusqu'à la défaillance, l'étude a maintenu une observation précise. Les enregistrements vidéo et l'analyse au microscope ont offert un examen détaillé, distinguant entre la déformation élastique et/ou plastique, élucidant les distorsions réversibles des altérations irréversibles induites par la force appliquée. L'inspection des spécimens (20x ; 50x ; VHX-6000, Keyence) a également contribué à une meilleure compréhension, documentant les fractures, les lignes de fissures visibles ou la déformation plastique au niveau de l'implant, du pilier et/ou de la couronne.

2.5. Limitations et perspectives de recherche future

Bien que cette étude fournisse des informations précieuses sur la stabilité mécanique des assemblages ti-base/iSC, certaines limitations qui pourrait influencer la possibilité de généralisation et l'interprétation des résultats doivent être reconnues.

1. Spécificité des matériaux : L'accent mis sur les iSC en PICN implique la nécessité de prendre des précautions lors de l'extrapolation de ces résultats à d'autres matériaux prothétiques. Les variations des

propriétés des matériaux, y compris le comportement mécanique sous contrainte, la force de liaison et l'usinabilité, peuvent affecter différemment la stabilité mécanique des assemblages ti-base/iSC pour différents matériaux.

2. Cadre in vitro : La nature in vitro de l'étude peut limiter sa capacité à reproduire parfaitement les conditions dynamiques rencontrées dans l'environnement oral. Bien qu'offrant des conditions contrôlées, l'absence de facteurs biologiques et les conditions in situ complexes nécessitent une attention particulière lors de l'application des résultats sur le plan clinique.
3. Conditions d'essai : Certains aspects de l'environnement de mastication simulé peuvent ne pas refléter complètement les véritables forces de mastication et la dynamique occlusale. Les variations dans l'application de la force, la direction et la fréquence in vivo pourraient influencer différemment la performance des assemblages ti-base/iSC.
4. Généralisation du protocole de scellement : Bien qu'insistant sur l'importance du protocole de scellement spécifique utilisé, la prudence est de mise lors de la généralisation de ces résultats à tous les scénarios de collage. Différents matériaux de restauration, ciments et procédures de scellement peuvent produire des résultats divers, nécessitant des approches sur mesure en milieu clinique.
5. Analyse des déformations : L'analyse des déformations sous charge statique fournit des informations précieuses, mais la distribution complexe des contraintes au sein des connexions implant-pilier peut impliquer des paramètres supplémentaires qui ne sont pas entièrement saisis. Une attention particulière est nécessaire lors de l'interprétation des implications des déformations observées.

Les futures recherches pourraient aborder ces limitations en incorporant divers matériaux prothétiques, en passant à des modèles in vivo et en explorant des corrélations cliniques plus vastes, renforçant ainsi la robustesse et l'applicabilité des résultats dans des scénarios cliniques réels.

En résumé, cette étude offre des perspectives sur les facteurs influençant la stabilité mécanique des assemblages ti-base/iSC. Les déformations observées sous charge statique suggèrent des implications potentielles de la conception de la connexion implant-pilier sur la distribution des contraintes. Cependant, il est crucial de noter que d'autres paramètres peuvent contribuer à ces résultats, nécessitant une interprétation prudente. De plus, l'étude s'est concentrée sur les iSC en PICN, et ces résultats ne s'appliquent peut-être pas directement à d'autres matériaux prothétiques, soulignant la nécessité d'une réflexion attentive et de recherches

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3. MANUSCRIT ORIGINAL

Title

The influence of titanium-base abutment geometry and height on mechanical stability of implant-supported single crowns

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Abstract

Aim: This study aimed to investigate the influence of titanium base (ti-base) abutment macro and micro geometry on the mechanical stability of polymer-infiltrated ceramic network (PICN) screw-retained implant-supported single crowns (iSCs).

Materials and Methods: Twelve specimens per group were used, comprising six different implant/ti-base abutment combinations restored with PICN iSCs: Nb-T (gingival height [GH]: 1.5 mm, prosthetic height [PH]: 4.3 mm), CC (GH: 0.8 mm, PH: 4.3 mm), CC-P (GH: 0.8 mm, PH: 7 mm), Nb-V (GH: 1.5 mm, PH: 6 mm), St (GH: 1.5 mm, PH: 5.5 mm), and Th (GH: 0.5 mm, PH: 9 mm). The specimens underwent thermo-mechanical aging, and those that survived were subsequently subjected to static loading until failure. The data was analysed using a one-way ANOVA test followed by Tukey post hoc test. ($\alpha = .05$)

Results: All specimens survived thermo-mechanical aging without complications namely visible cracks, debonding, or screw loosening. Th group demonstrated the highest strength values among all the groups, with significant differences compared to Nb-T ($p < .05$), CC ($p < .001$), and St ($p < .001$). Additionally, CC-P group exhibited significantly superior fracture strength results compared to CC ($p < .05$) and St ($p < .05$).

Conclusion: The choice of ti-base, particularly prosthetic height had a significant influence on fracture resistance of PICN iSCs. Nevertheless, the height or geometrical features of the ti-base did not exhibit a significant influence on the mechanical behaviour of the iSC/ti-base assembly under thermomechanical loading, as all specimens withstood the aging without complication or failure.

Keywords: crowns, implant abutments, dental implants, fracture load, hybrid ceramic, polymer-infiltrated-ceramic-matrix, PICN

1. Introduction

The use of titanium base (ti-base) abutments has significantly enhanced the prosthetic options available for implant-supported restorations, providing clinicians with a more versatile approach to improve prosthetic outcomes. Ti-base abutments provide a stable implant abutment connection and allow for a customized emergence profile without the added cost of custom abutments, as they are prefabricated. Moreover, their easy retrievability is an additional advantage for clinicians.

However, the mechanical stability of the crown/ti-base assembly largely depends on the strength of the interface between the two components, namely the retention (Rödiger, Kloß, Gersdorff, Bürgers, & Rinke, 2018) and adhesion aspects (Ioannidis et al., 2020; Zahoui et al., 2020). Previous studies have investigated the influence of surface treatments (Pitta, Burkhardt, et al., 2021), adhesive protocols (Burkhardt, Sailer, Fehmer, Mojon, & Pitta, 2022; Zahoui et al., 2020), and their relationship with different prosthetic materials (Burkhardt et al., 2022) on the mechanical stability and pull-off forces of the implant-supported single crowns (iSCs) (Att, Hoischen, Gerds, & Strub, 2008). It was found that a surface treatment by means of sand-blasting is necessary to improve the bonding forces and thus the mechanical stability of iSC to the ti-base abutment (Pitta, Burkhardt, et al., 2021). Furthermore, the use of the correct adhesive protocol according to the restorative material selected was reported to be essential in achieving sufficient pull-off forces (Burkhardt et al., 2022; Lopes et al., 2019).

Ti-bases have different macro and micro geometry factors, such as abutment prosthetic height (PH), ti-base adhesive surface, antirotational features of the ti-base, diameter of the prosthetic part, surface characteristics that might influence the bonding stability between the iSCs and the ti-base. Although there has been a considerable amount of research on surface treatments and bonding protocols for ti-base abutments, few studies have focused on the impact of ti-base geometry which are predominantly in vitro pull-off tests (Rödiger et al., 2018; Silva et al., 2018; Zahoui et al., 2020). The impact of abutment macro and micro geometry on the overall mechanical stability of the crown-abutment assembly remains largely unexplored. As such, further studies are necessary to explore the relationship between the mentioned macro and micro geometry features and mechanical stability of the iSCs which can aid in optimizing the ti-base design, abutment selection and accordingly clinical success of ti-base abutment supported iSCs (Chantler et al., 2023; Derksen et al., 2023).

Therefore, this in vitro study aimed to evaluate the combined influence of abutment macro and micro geometry factors by testing 6 different ti-base abutment/implant combinations on the mechanical stability of screw-retained iSCs made of polymer-infiltrated-ceramic-network (PICN). Accordingly, the null hypothesis for this study is “abutment geometry would not have an effect on the overall mechanical stability of implant-supported screw-retained single crowns with ti-base abutments.”

2. Materials and Methods

2.1. Study setting

Twelve specimens per group were included in this study, a number which was based on a previous similar study (Pitta, Hjerpe, et al., 2021). Six study groups comprised of specimens fabricated using 6 different implant/ti-base abutment combinations restored with screw retained polymer-infiltrated ceramic network (PICN) crowns (VITA Enamic, VITA Zahnfabrik). The study groups and their implant, abutment, and specimen manufacturing details are presented in Table 1 and Figure 1. In five out of six groups, commercially available ti-base abutments were used. However, in one group (CC-P), prototype ti-base abutments with a larger prosthetic height compared to their commercially available counterparts (CC) were specifically manufactured by the manufacturers for the present study. This was done to facilitate a direct comparison of two different ti-base heights (Table 1 and Figure 1).

2.2. Fabrication of restorations

A maxillary Typodont model (Frasaco GmbH) was modified by removing the right 2nd premolar, and the space was filled with wax. The model was then scanned (Iscan D103i, Imetric 3D SA) and six 3D printed master models (SheraPrint Model Plus Sand, SHERA Werkstoff-Technologie GmbH & Co & RapidShape P30, Rapidshape, Straumann) were obtained. On these models, implant holes for each planned implant type and diameter (Table 1) were created using the corresponding drilling protocol. The implants were then placed according to the prospective crown position. Scanbodies were mounted on the implants (Table 1) and digital impressions from the master models with the respective implants and scanbodies were taken using an intraoral scanner (3 Shape Trios 3, 3 Shape) except for CC-P group, including a prototype ti-base (Conelog, Biohorizons Camlog). As this prototype ti-base, specifically designed and fabricated for the present study, was not existing

in the design software library used (Dental System, version 2018, 3 Shape); an abutment level impression instead of implant level impression was taken for the specimens with prototype ti-base abutment. The crown height (from ti-base platform to the crown buccal cusp tip) was defined as 10.5 mm which was 1.5 times higher than a mean reported premolar height (Magne, Gallucci, & Belser, 2003) with a purpose of creating a clinical scenario that can require an increased crown height. The implant crowns were then designed (Dental System, version 2018, 3 Shape) with the same outer shape and dimensions for all groups (Table 1), and milled (CEREC MC XL, Dentsply Sirona) out of the PICN ingots (VITA Enamic IS-16S—HT 2M2, VITA Zahnfabrik).

2.3. Specimen fabrication

ISO Standard 14801 recommendations (“Dentistry-Implants-Dynamic fatigue test for endosseous dental implants,” International Organization for Standardization 2016, Geneva, Switzerland) were followed and a total number of 72 implants (Table 1) were embedded into custom-made plexiglass holders using a self-curing acrylic (Technovit 4,071, Kulzer GmbH). According to the ISO standard, the implants were embedded in the blocks with the 3 mm neck portion exposed to simulate vertical bone loss. The ti-base abutments were screw-retained to their respective implants with their corresponding recommended manufacturer torques (Table 1, Figure 1).



Figure 1. The study groups and ti-base/implant combinations used for the study. From left to right: Nb-T (Nobel Biocare Universal Base, gingival height [GH]: 1.5 mm, prosthetic height [PH]: 4.3 mm & Nobel Biocare), CC (Camlog Conelog GH: 0.8 mm, PH: 4.3 mm), CC-P (Camlog Conelog prototype titanium base abutment GH: 0.8 mm, PH: 7 mm), Nb-V (Ivoclar Viteobase GH: 1.5 mm, PH: 6 mm), St (Straumann Variobase GH: 1.5 mm, PH: 5.5 mm), and Th (Thommen Varioflex GH: 0.5 mm, PH: 9 mm)

Group	Abutment Properties			Implant Properties			Crown Properties		Cementation protocol					
	Brand	Type	Gingival Height [mm]	Prosthetic Height [mm]	Torque value [N/cm]	Brand	Type	Diameter (ø) & Length [mm]	Implant Abutment connection	Material & Brand	Dimensions	Ti base Surface Treatment	Crown Surface treatment	Cement
Nb-T (n=12)	Nobel Biocare	Universal Base, Crown	1.5	4.3	35	Nobel Biocare	NobelActive	ø 4.3, 13	Internal-conical connection					
CC (n=12)	Biohorizons Camlog	Coneolog Titanium Base, CAD/CAM crown	0.8	4.3	20	Biohorizons Camlog	Coneolog	ø 4.3, 13	Internal-conical connection					
CC-P (n=12)	Biohorizons Camlog	Prototype Coneolog Titanium Base, CAD/CAM crown	0.8	7	20	Biohorizons Camlog	Coneolog	ø 4.3, 13	Internal-conical connection	Polymer Infiltrated Ceramic Network (PICN), VITA Enamic	10.5 Height, 7.9 MD, 9.5 BL	50-µm Al ₂ O ₃ sand-blasting under 2.5 bars 40 angle for 20 sec/cm ² (except Viteobase ti-base), Ultrasonic bath with Alcohol (4 mins), Clearfil primer 20 sec application and air dry	Acid etching (5% HCl acid, 60 sec), Ultrasonic cleaning (4 mins with alcohol), Clearfil primer (20secs application/air dry)	Panavia V5 application on both crown inner surface and ti-base surface
Nb-V (n=12)	Ivoclar Vivadent	Viteobase, Crown	1.5	6	35	Nobel Biocare	NobelActive	ø 4.3, 13	Internal-conical connection					
St n=12)	Straumann	Variobase	1.5	5.5	35	Straumann	Bone Level	ø 4.1, 12	Internal-conical connection					
Th (n=12)	Thommen Medical	Varioflex, Crown	0.5	9	25	Thommen Medical	Thommen Element	ø 4.3, 14	Butt-joint connection					

Table 1. Material, manufacturing, and adhesive protocol details of study groups.

GH: gingival height;
PH: prosthetic height;
MD: Mesiodistal;
BL: Buccolingual

The abutment and crown bonding surfaces had been previously treated according to a previously established protocol (Burkhardt et al., 2022). The protocol was adopted from a previous study (Burkhardt et al. 2022) that showed the most favourable result for bonding the PICN crowns to ti-base abutments. Briefly, the PICN crowns were etched with 5% HF (IPS ceramic etching gel, Ivoclar Vivadent AG) for 60 s (Figure 2E); the ti-bases were sand-blasted (except for Viteobase ti-bases) using 50- μm Al_2O_3 (Figure 2A, 2B) particles for 20 s/ cm^2 at 2.5 bars (Burkhardt et al. 2022) (Figure 2). After surface treatment, crowns and abutments were cleaned in an ultrasonic alcohol bath (Micro 10+, Unident) for 4 min, and then a 10-methacryloyloxydecyl dihydrogen phosphate (MDP) monomer-containing primer was applied for both ceramic and metallic surfaces (Clearfil Ceramic Primer Plus, Kuraray) (Figure 2C, 2F) (Figure 3). The crowns then were bonded to the ti-bases with a MDP monomer-containing resin cement (Panavia V5, Kuraray) (Figure 2H), after the respective surface pre-treatments.

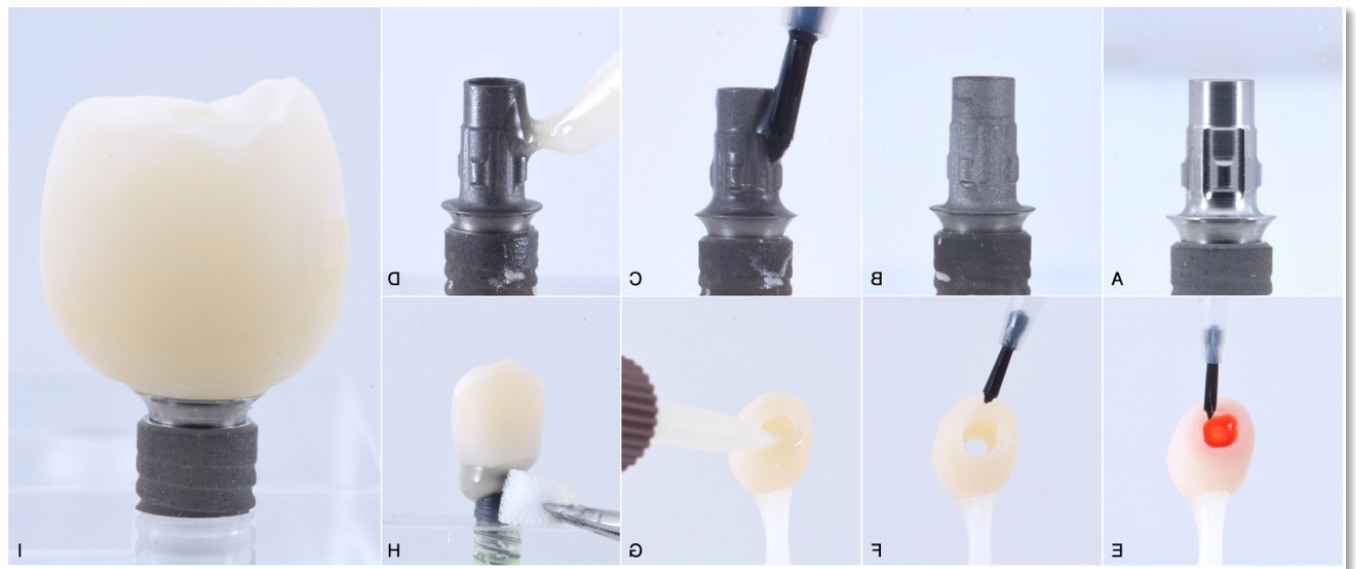


Figure 2. Adhesive protocol that was followed for bonding of PICN iSCs to ti-base abutments.

A. Representative specimen from St group; B. 50- μm Al_2O_3 sand-blasting under 2.5 bars 20 sec/ cm^2 and ultrasonic cleaning (4 min with alcohol); C. primer application (20 s) and air dry; D. resin cement application; E. Acid etching (5% HCl acid, 60 s) and ultrasonic cleaning (4 min with alcohol); F primer application (20 sec application/air dry); G. Resin cement application; H. excess removal; I. representative specimen from St group after cementation of PICN iSC.



Figure 3. 10-methacryloyloxydecyl dihydrogen phosphate (MDP) monomer-containing primer was applied for both ceramic and metallic surfaces (Clearfil Ceramic Primer Plus, Kuraray)

The screw access holes were closed with polytetrafluoroethylene (PTFE) tape (Debrunner Acifer AG) and a resin based composite filling material (Tetric EvoCeram, Ivoclar Vivadent) (Figure 4).



Figure 4. Study groups specimens after cementation.

2.4. Thermo-mechanical aging

A simultaneous thermocycling (5°–55°C, dwelling time 120 s) and chewing simulation (1,200,000 cycles, 98 N 1.67 Hz, CS-4.8, SD Mechatronik) was applied to all specimens. The specimens were loaded from the occlusal surface 2 mm lingual from the screw access channel parallel to the implant axis, and with a vertical movement of 2 mm. Ceramic balls with Ø 6 mm (Steatite ball, Hoechst CeramTec) that approximates the

Vickers hardness of enamel were fixed to the indenters to be used as antagonists.

2.5. Complications and outcome measures

After the thermo-mechanical aging, all surviving specimens underwent thorough examination to identify and classify non-catastrophic events such as cracks, debonding, or screw loosening. Two calibrated observers (DK, JP) evaluated these events using a light microscope with 50× magnification (Olympus SZX9, Olympus). The outcome measures to identify non-catastrophic events were as following:

- Debonding incidents: micro-movement : when slight movement between the abutment and the crowns is visible under a microscope and loss of retention; macro-movement : the movement is also visible to the naked eye ; loss of retention : the complete displacement of the crown from the abutment.
- Screw loosening : any movement between the abutment and the implant.

2.6. Fracture load test

The survived specimens were statically loaded parallel to the implant axis until failure using a universal testing machine (Shimadzu AGS-X series, Shimadzu). A 0.5 mm thick tin foil (Dentaurum GmbH) was placed between the specimens and the indenter, to ensure even force distribution to the crown/abutment during loading. Failure was defined either as visible fracture of the abutment/ crown, or as 20% decrease of the maximum load during the fracture load test if no fracture was detected. The fracture load of each group was recorded using specific software (TRAPEZIUM X, V.1.4.4, Shimadzu).

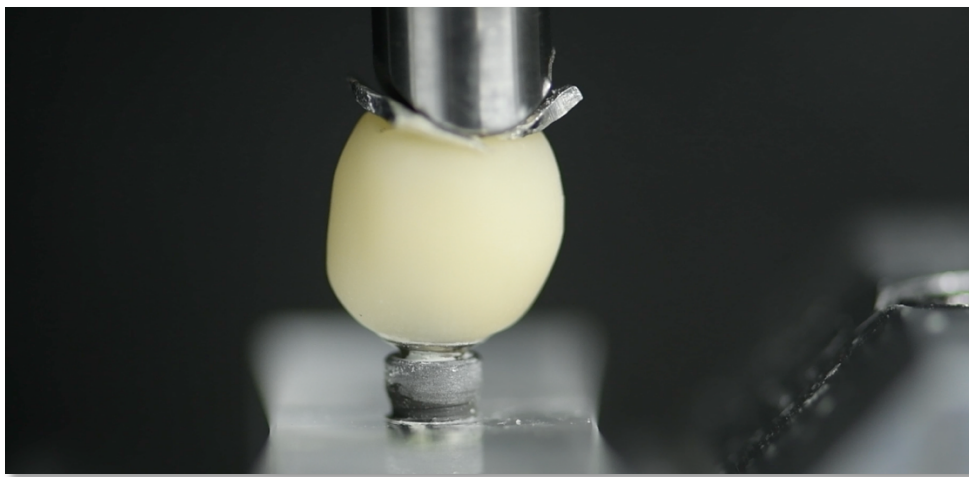


Figure 5. Fracture load testing (Shimadzu AGS-X series, Shimadzu).

2.7. Failure analysis

During the fracture load test video recordings with magnification (Nikon D250 and Macro Lens) were done, the recordings were later analyzed to detect if any elastic and/or plastic deformation had happened during the test. Elastic deformation was defined as the reversible distortion of implant-abutment assembly in response to applied force, where the material returned to its original shape once the force is removed whereas plastic deformation was defined as the irreversible alteration of the components resulting in permanent changes. The specimens then were inspected (20x; 50x; VHX-6000, Keyence) for determining the mode of failure. Failures, if any, including fractures, visible crack lines, or plastic deformation at implant, abutment, and/or crown level were documented.

An additional scanning electron microscope (SEM) analysis on the selected specimens from each group were done to identify the failure characteristics using an SEM (11x, 25x, 50x, and 200x, Gemini Zeiss Sigma 300 VP; Carl Zeiss) (Figure 6A-F).



Figure 6. SEM images from the representative specimens from study groups. **A.** Nb-T; **B.** CC; **C.** CC-P; **D.** Nb-V; **E.** St **F.** Th

2.8. Statistical Analysis

Shapiro-Wilk tests were used to verify the normal distribution of the mean fracture load values across ti-base abutment types. All study groups showed normal distribution, therefore a parametric test, one way ANOVA, was performed and followed by Tukey post hoc test to detect the difference across the study groups. The elastic deformations occurrence during load to fracture test was compared by Pearson's chi-square tests. The level of significance was set at $\alpha = .05$ for all the tests. The statistical analysis was performed using a statistical software program (IBM SPSS Statistics, v24.0; IBM Corp).

3. Results

All specimens survived the thermomechanical aging resulting in 100% survival rate for all groups after 1.200.000 cycles. No visible cracks were observed in any of the implant components or crowns, nor were crown debonding and screw loosening detected. The evaluation of the specimens under magnification presented no micro-movement due to debonding.

The descriptive statistics for the fracture strength values are presented in Table 2.

Table 2. Fracture strength test results.

Groups	Fracture Strength						Deformation Ratio
	N	Mean (N)	SD	Median (N)	SE	Turkey test* ($\alpha = .05$)	
Nb	12.00	2231.43	582.90	2097.86	168.27	a, b	0.58
CC	12.00	2029.97	278.87	2026.17	80.50	a	0.75
CC-P	12.00	2554.78	408.03	2442.02	117.79	b, c	0.49
Nb-V	12.00	2316.93	348.90	2362.84	100.72	a, b, c	0.83
St	12.00	2013.92	329.23	1994.49	95.04	a	0.33
Th	12.00	2764.43	477.03	2691.68	137.71	c	0.00

SD: Standard Deviation; SE: Standard error; N: Newton; Nb-T (Nobel Biocare Universal Base);

CC (Camlog Conelog); CC-P (Camlog Conelog prototype); Nb-V (Ivoclar Viteobase), St

(Straumann Variobase); and Th (Thommen Varioflex)

*Different small letters identifies the significant difference ($\alpha = .05$)

One-way ANOVA test detected a significant difference between the groups ($F(5, 66) = 6.06$ $p < .001$), therefore followed by Tukey post-hoc test. CC-P group showed significantly better fracture strength than CC ($p < .05$) and St ($p < .05$). Furthermore, Th group showed highest strength values among the groups with significant differences compared to Nb-T ($p < .05$), CC ($p < .001$), and St ($p < .001$) groups (Table 2 and Figure 7).

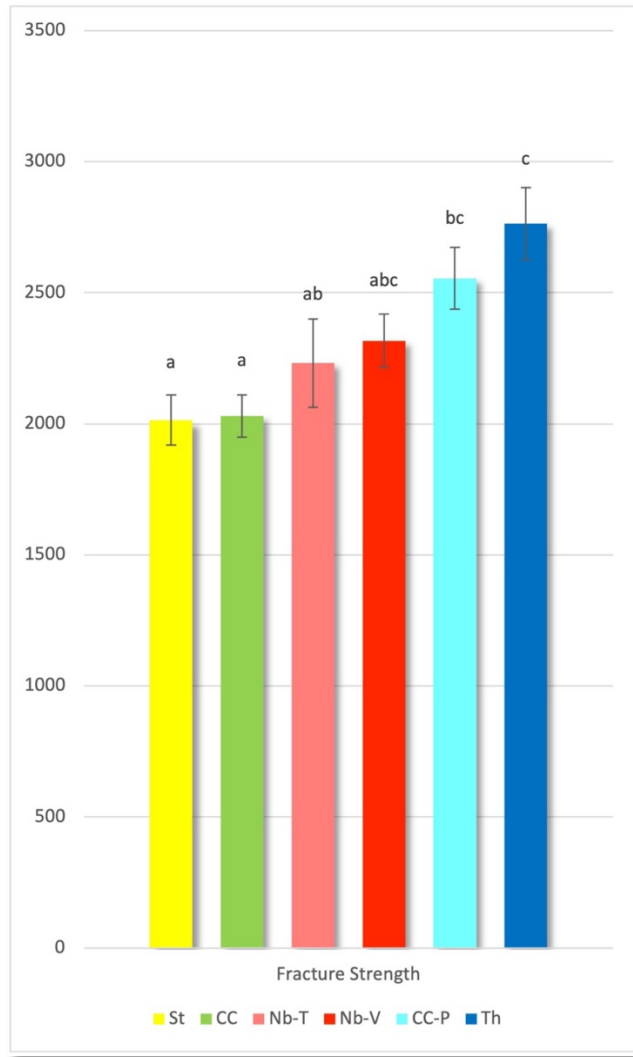


Figure 7. Fracture resistance values and Tukey post-hoc test results (Different letters indicates significance difference between groups $\alpha = 0.05$)

A chi-squared test was applied to find any difference across the groups regarding deformation under static loading (Table 2). The test indicated a difference was found across the groups ($\chi^2 = 23.18, p < .001$) (Table 2). The deformation was prominent for CC-P and Nb-V groups which were accompanied by conical connection ti-base abutments with biggest height and biggest circumferential dimensions, respectively. The deformation observed was of a plastic nature, manifesting as a bending of the implant-iSC assembly towards the lingual, where the static-axial loading was applied. No elastic or plastic deformation under static loading was detected among Th group specimens, which was the only study group with butt-joint implant-abutment connection. The fracture started from the cervical area of the PICN crowns and tracted towards occlusal plane by creating a clear fracture line in majority of the specimens (Figure 8A-F) The fractographic analysis conducted using SEM revealed that there were no notable distinctions in the fracture patterns observed among various ti-base designs (Figure 6A-F).

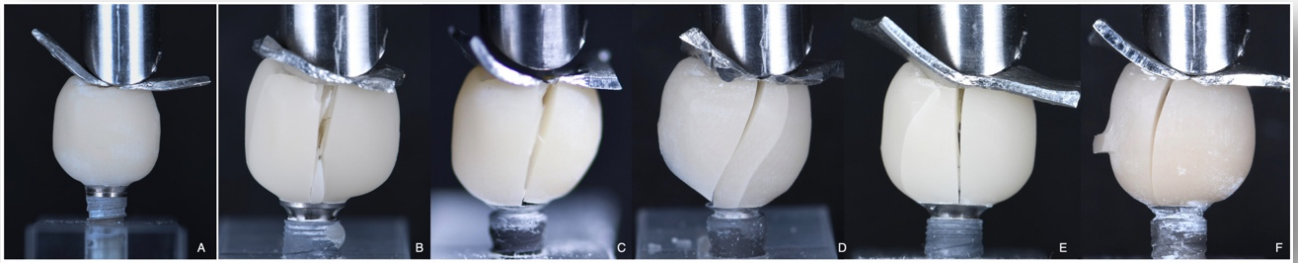


Figure 8. static loading in a universal testing machine and representative images from study groups **A.** Th **B.** Nb-T; **C.** CC; **D.** CC-P; **E.** Nb-V; **F.** St

4. Discussion

The results of the present study showed that prosthetic height of the ti-base significantly influenced the fracture strength of the iSCs. However, other abutment features, such as pre-packaging sand-blasting of the titanium base surface or the presence of antirotational macro-geometrical features did not demonstrate a significant effect. Importantly, no complications related to ti-base, iSC, or implant were observed during the thermomechanical aging test. Therefore, the null hypothesis was partially rejected.

All specimens survived the thermomechanical aging process without any visible cracks, crown debonding, screw loosening, or debonding-related micro-movement, demonstrating a 100% survival rate. The specimens were loaded with 98N, which exceeds the expected maximum chewing force of 70N occurring in the molar area in the absence of parafunctional activity (Bates, Stafford, & Harrison, 1976; De Boever, McCall, Holden, & Ash, 1978; Kohyama et al., 2004; Schindler, Stengel, & Spiess, 1998).

Similarly, in a study by Yazigi et al. (2020), which investigated various iSC materials, including zirconia ceramic, lithium disilicate, nano-hybrid composite resin, and PICN, under similar thermocycling conditions and a 49N loading force, no debonding or mechanical failures were observed during thermomechanical loading. However, it is noteworthy that the same study reported median fracture resistance values of 670N (IQR: 111) for PICN iSCs, which were significantly lower than the findings of the present study and previous data published on PICN fracture resistance (Yazigi, Kern, Chaar, Libeck, & Elsayed, 2020). The test settings in Yazigi et al. (2020) 's study were similar to the present study, as both investigations included

thermomechanical aging prior to the fracture resistance test, and the loading was applied parallel to the axial axis of the implant (Yazigi et al., 2020). However, one notable difference was the utilized an occlusal load which was set to 49 N by Yazigi et al. whereas present study used a higher occlusal load of 100 N. Nonetheless, it is important to mention that Yazigi et al. did not provide information about the mode of failure or the fractographic analysis of the specimens, which could have further elucidated the reasons behind the observed differences in fracture resistance values (Yazigi et al., 2020).

On the other hand, Sieper et al. investigated the fracture behaviour of PICN and lithium disilicate molar crowns, reporting significantly lower mean fracture strength values for PICN compared to lithium disilicate (Sieper, Wille, & Kern, 2017). However, the reported mean fracture strength values for PICN crowns after artificial aging were 2128 N (1 mm thickness) and 2281 N (1.5 mm thickness), both of which exceed physiological masticatory forces (Sieper et al., 2017). These results are consistent with the present study.

Undoubtedly, the high success rate observed after the thermocycling test in the present study can be attributed to the bonding protocol followed. Previous research has consistently emphasized the significance of surface treatments (Pitta, Burkhardt, et al., 2021) and bonding protocols (Burkhardt et al., 2022) in enhancing the mechanical stability of the iSC/ti-base assembly. In the current study, the adhesive protocol adopted from Burkhardt et al., which yielded the best results for bonding PICN iSCs to ti-base abutments. Burkhardt et al. (2022) investigated the bonding of three different restorative materials (PICN, Lithium Disilicate, and zirconia) to ti-bases using three different resin cements (Multilink Hybrid abutment, Panavia V5, and RelyX Ultimate), along with their corresponding primers. Notably, when PICN iSCs were combined the MDP monomer-containing cement system selected in the present study (Clearfil Ceramic Primer Plus and Panavia V5, Kuraray), they exhibited significantly less debonding compared to the other two restorative materials (Burkhardt et al., 2022). The adoption of this specific bonding protocol has contributed to the superior performance of the PICN iSCs in our study, further solidifying the importance of using appropriate resin cements tailored to specific restorative materials for achieving optimal bonding and mechanical stability.

The current study's results also suggest that ti-base prosthetic height is a significant factor in the fracture resistance of the crown-titanium base assembly. Longer abutments (CC-p: prosthetic height (PH)=7; Th:

PH=9) had a significantly higher fracture strength force than shorter abutments (CC: PH=4.3 mm and Nb-T: PH=4.3). Rodiger et al. (2018) investigated the influence of ti-base length as well as the adhesive cement used on the pull-off forces of single implant crowns on ti-bases (Rödiger et al., 2018). In parallel to the present study's findings, increased height of the abutment resulted in a significant increase in the removal forces (Rödiger et al., 2018).

Interestingly, the height of the titanium base had a greater impact on fracture resistance compared to other properties examined, such as pre-packaging sand-blasting of the titanium base surface (Nb-V, Figure 1) or the presence of antirotational macro-geometrical features (such as in Nb-T and St groups, Figure 1) (Table 2). Moreover, neither the anti-rotational features nor the prosthetic height (PH) of the abutment demonstrated influence on mechanical stability, as all specimens withstood the 1.200.000 cycle of thermomechanical loading.

The fracture loads observed in each group of specimens exceeded 2000N, significantly surpassing the typical masticatory force range of 346-548 N (Karasan, Canay, Sailer, & Att, 2022). This high strength can be attributed, in part, to the mechanical properties of the PICN material. Research suggests that PICN materials exhibit improved fracture toughness and machinability compared to ceramics, due to their reduced brittleness, rigidity, and hardness, coupled with enhanced flexibility (Coldea, Swain, & Thiel, 2013).

Moreover, the application of a hybrid ceramic material with moderately elastic characteristics in iSCs showed a significant capacity to absorb or diminish around 70% of the masticatory forces transmitted to the simulated periimplant bone (Menini et al., 2013). This notable improvement stands in contrast to the stiffness typically associated with materials like zirconia, glass ceramic, and gold. These findings also can explain the 100% survival rate and absence of complications observed during the thermomechanical aging test.

Based on the observed deformations under static loading, it can be suggested that the implant-abutment connection design may influence the distribution of stresses. Notably, the butt-joint implant abutment-connection displayed no deformation under load, suggesting its potential to distribute stress more effectively and minimize its transfer to the implant. In contrast, abutments with larger circumferential dimensions and robust structures were associated with plastic deformation and bending of the implant which might be

explained by greater stress transfer to the implant. While the observed deformations offer valuable insights, it's important to recognize that other parameters may also impact these results. Therefore, drawing conclusions about the causality of deformation patterns and the influence of implant-abutment connections requires careful consideration.

It is important to emphasize that this study examined PICN iSCs, and its findings may not directly apply to other prosthetic materials. The mechanical behaviour under stress, bonding strength, the influence of chosen bonding protocol, and machinability can differ among various prosthesis materials. These disparities could potentially affect the mechanical stability of the ti-base/iSC assembly. (Patzelt, Spies, & Kohal, 2015). This is why, standardizing the iSC material was essential to ensure the standardization among the study groups in order to investigate the influence of ti-base geometry. Moreover, the study was conducted in vitro, and the findings should be interpreted with caution.

5. Conclusions

Within the scope and limitations of this study, the selection of the ti-base abutment demonstrated a significant influence on the fracture resistance of screw-retained PICN iSCs. Notably, the height of the titanium base was the predominant factor impacting fracture resistance, superseding the influence of other geometric features namely anti-rotational features and pre-packaging sandblasting. Interestingly, ti-base abutment characteristics showed no discernible effect on mechanical stability during the aging, as all specimens withstood the thermomechanical loading without complication or failure. Further research is necessary to provide recommendations for optimal ti-base geometry that enhances mechanical stability and prolongs the longevity of implant-supported restorations.

Acknowledgements

The authors would like to thank to Prof Dr Susan Scherrer from University of Geneva, University Clinics of Dental Medicine, Division of Fixed Prosthodontics and Biomaterials for her support in SEM analysis and for providing SEM images. An acknowledgment is also given to the manufacturers Biohorizons Camlog,

Straumann, Nobel Biocare (Grant no: 2019-1615), Thommen Medical, Ivoclar Vivadent, Vita Zahnfabrik and for their support with the respective materials.

Conflict of Interest

Authors declare no conflict of interest.

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FIGURE LEGENDS

Figure 1. The study groups and ti-base/implant combinations used for the study. From left to right: Nb-T (Nobel Biocare Universal Base, gingival height [GH]: 1.5 mm, prosthetic height [PH]: 4.3 mm & Nobel Biocare), CC (Camlog Conelog GH: 0.8 mm, PH: 4.3 mm), CC-P (Camlog Conelog prototype titanium base abutment GH: 0.8 mm, PH: 7 mm), Nb-V (Ivoclar Viteobase GH: 1.5 mm, PH: 6 mm), St (Straumann Variobase GH: 1.5 mm, PH: 5.5 mm), and Th (Thommen Varioflex GH: 0.5 mm, PH: 9 mm)

Figure 2. Adhesive protocol that was followed for bonding of PICN iSCs to ti-base abutments.

A. Representative specimen from St group; B. 50- μm Al_2O_3 sand-blasting under 2.5 bars 20 sec/ cm^2 and ultrasonic cleaning (4 min with alcohol); C. primer application (20 s) and air dry; D. resin cement application; E. Acid etching (5% HCl acid, 60 s) and ultrasonic cleaning (4 min with alcohol); F primer application (20 sec application/air dry); G. Resin cement application; H. excess removal; I. representative specimen from St group after cementation of PICN iSC.

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Figure 8. Fracture resistance values and Tukey post-hoc test results (Different letters indicates significance difference between groups $\alpha = 0.05$)

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ANNEXES

Article original publié dans Clinical Oral Implants Research

Karasan D, Pitta J, Zarauz C, Strasding M, Liu X, Fehmer V, Sailer I. The influence of titanium-base abutment geometry and height on mechanical stability of implant-supported single crowns. Clin Oral Implants Res. 2023 Nov 6. doi: 10.1111/clr.14207. Epub ahead of print. PMID: 37929698.

<https://pubmed.ncbi.nlm.nih.gov/37929698/>

The influence of titanium-base abutment geometry and height on mechanical stability of implant-supported single crowns

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Abstract

Aim: This study aimed to investigate the influence of titanium base (ti-base) abutment macro- and micro-geometry on the mechanical stability of polymer-infiltrated ceramic network (PICN) screw-retained implant-supported single crowns (ISCs).

Materials and Methods: Twelve specimens per group were used, comprising six different implant/ti-base abutment combinations restored with PICN ISCs: Nb-T (gingival height [GH]: 1.5 mm, prosthetic height [PH]: 4.3 mm), CC (GH: 0.8 mm, PH: 4.3 mm), CC-P (GH: 0.8 mm, PH: 7 mm), Nb-V (GH: 1.5 mm, PH: 6 mm), St (GH: 1.5 mm, PH: 5.5 mm), and Th (GH: 0.5 mm, PH: 9 mm). The specimens underwent thermo-mechanical aging, and those that survived were subsequently subjected to static loading until failure. The data were analyzed using a one-way ANOVA test followed by Tukey post hoc test ($\alpha = .05$).

Results: All specimens survived thermo-mechanical aging without complications, namely, visible cracks, debonding, or screw loosening. Th group demonstrated the highest strength values among all the groups, with significant differences compared to Nb-T ($p < .05$), CC ($p < .001$), and St ($p < .001$). Additionally, CC-P group exhibited significantly superior fracture strength results compared to CC ($p < .05$) and St ($p < .05$).

Conclusion: The choice of ti-base, particularly prosthetic height, had a significant influence on fracture resistance of PICN ISCs. Nevertheless, the height or geometrical features of the ti-base did not exhibit a significant influence on the mechanical behavior of the ISC/ti-base assembly under thermomechanical loading, as all specimens withstood the aging without complication or failure.

KEYWORDS

crowns, dental implants, fracture load, hybrid ceramic, implant abutments, PICN, polymer-infiltrated-ceramic-matrix

1 | INTRODUCTION

The use of titanium base (ti-base) abutments has significantly enhanced the prosthetic options available for implant-supported restorations, providing clinicians with a more versatile approach to

improve prosthetic outcomes. Ti-base abutments provide a stable implant abutment connection and allow for a customized emergence profile without the added cost of custom abutments, as they are prefabricated. Moreover, their easy retrievability is an additional advantage for clinicians.

using 6 different implant/ti-base abutment combinations restored with screw retained polymer-infiltrated ceramic network (PICN) crowns (VITA Enamic, VITA Zahnfabrik). The study groups and their implant, abutment, and specimen manufacturing details are presented in Table 1 and Figure 1. In five out of six groups, commercially available ti-base abutments were used. However, in one group (CC-P), prototype ti-base abutments with a larger prosthetic height compared to their commercially available counterparts (CC) were specifically manufactured by the manufacturers for the present study. This was done to facilitate a direct comparison of two different ti-base heights (Table 1 and Figure 1).

2.2 | Fabrication of restorations

A maxillary Typodont model (Frasaco GmbH) was modified by removing the right 2nd premolar, and the space was filled with wax. The model was then scanned (Iscan D103i, Imetric 3D SA), and six 3D printed master models (SeraPrint Model Plus Sand, SHERA Werkstoff-Technologie GmbH & Co & RapidShape P30, Rapidshape, Straumann) were obtained. On these models, implant holes for each planned implant type and diameter (Table 1) were created using the corresponding drilling protocol. The implants were then placed according to the prospective crown position.

Scanbodies were mounted on the implants (Table 1), and digital impressions from the master models with the respective implants and scanbodies were taken using an intraoral scanner (3 Shape Trios 3, 3 Shape) except for CC-P group, including a prototype ti-base (Conelog, Biohorizons Camlog). As this prototype ti-base, specifically designed and fabricated for the present study, was not existing in the design software library used (Dental System, version 2018, 3 Shape) an abutment level impression instead of implant level impression was taken for the specimens with prototype ti-base abutment. The crown height (from ti-base platform to the crown buccal cusp tip) was defined as 10.5 mm, which was 1.5 times higher than a mean reported premolar height (Magne et al., 2003) with a purpose of creating a clinical scenario that can require an increased crown height. The implant crowns were then designed (Dental System, version 2018, 3 Shape) with the same outer shape and dimensions for all groups (Table 1) and milled (CEREC MC XL, Dentsply Sirona) out of the PICN ingots (VITA Enamic IS-16S-HT 2 M2, VITA Zahnfabrik).

2.3 | Specimen fabrication

ISO Standard 14,801 recommendations ("Dentistry-Implants-Dynamic fatigue test for endosseous dental implants,"

Implant abutment connection	Crown properties		Cementation protocol		
	Material & brand	Dimensions	Ti base surface treatment	Crown surface treatment	Cement
Internal-connical connection	Polymer Infiltrated Ceramic Network (PICN), VITA Enamic	10.5 Height, 7.9 MD, 9.5 BL	50- μ m Al ₂ O ₃ sand-blasting under 2.5 bars 40 angle for 20 s/cm ² Ultrasonic bath with Alcohol (4 min), Clearfil primer 20 s application and air dry	Acid etching (5% Hydrofluoric acid, 60 s), Ultrasonic cleaning (4 min with alcohol), Clearfil primer (20 s application/air dry)	Panavia V5 application on both crown inner surface and ti-base surface
Internal-connical connection	as above	as above	as above	as above	as above
Internal-connical connection	as above	as above	as above	as above	as above
Internal-connical connection	as above	as above	N/A	as above	as above
Internal-connical connection	as above	as above	as above	as above	as above
Butt-joint connection	as above	as above	as above	as above	as above

International Organization for Standardization 2016, Geneva, Switzerland) were followed, and a total number of 72 implants (Table 1) were embedded into custom-made plexiglass holders using a self-curing acrylic (Technovit 4071, Kulzer GmbH). According to the ISO, the implants were embedded in the blocks with the 3 mm neck portion exposed to simulate vertical bone loss. The ti-base abutments were screw-retained to their respective implants with their corresponding recommended manufacturer torques (Table 1; Figure 1).

The abutment and crown bonding surfaces had been previously treated according to a previously established protocol (Burkhardt et al., 2022). The protocol was adopted from a previous study (Burkhardt et al., 2022) that showed the most favorable result for

bonding the PICN crowns to ti-base abutments. Briefly, the PICN crowns were etched with 5% HF (IPS ceramic etching gel, Ivoclar Vivadent AG) for 60s (Figure 2e); the ti-bases were sand-blasted (except for Viteobase ti-bases) using 50- μ m Al_2O_3 (Figure 2a,b) particles for 20s/cm² at 2.5 bars (Burkhardt et al., 2022) (Figure 2). After surface treatment, crowns and abutments were cleaned in an ultrasonic alcohol bath (Micro 10+, Unident) for 4 min, and then a 10-methacryloyloxydecyl dihydrogen phosphate (MDP) monomer-containing primer was applied for both ceramic and metallic surfaces (Clearfil Ceramic Primer Plus, Kuraray) (Figure 2c,f). The crowns then were bonded to the ti-bases with a MDP monomer-containing resin cement (Panavia V5, Kuraray) (Figure 2h), after the respective surface pre-treatments.



FIGURE 1 The study groups and ti-base/implant combinations used for the study. From left to right: Nb-T (Nobel Biocare Universal Base, gingival height [GH]: 1.5 mm, prosthetic height [PH]: 4.3 mm), CC (Camlog Conelog GH: 0.8 mm, PH: 4.3 mm), CC-P (Camlog Conelog prototype titanium base abutment GH: 0.8 mm, PH: 7 mm), Nb-V (Ivoclar Viteobase GH: 1.5 mm, PH: 6 mm), St (Straumann Variobase GH: 1.5 mm, PH: 5.5 mm), and Th (Thommen Varioflex GH: 0.5 mm, PH: 9 mm).

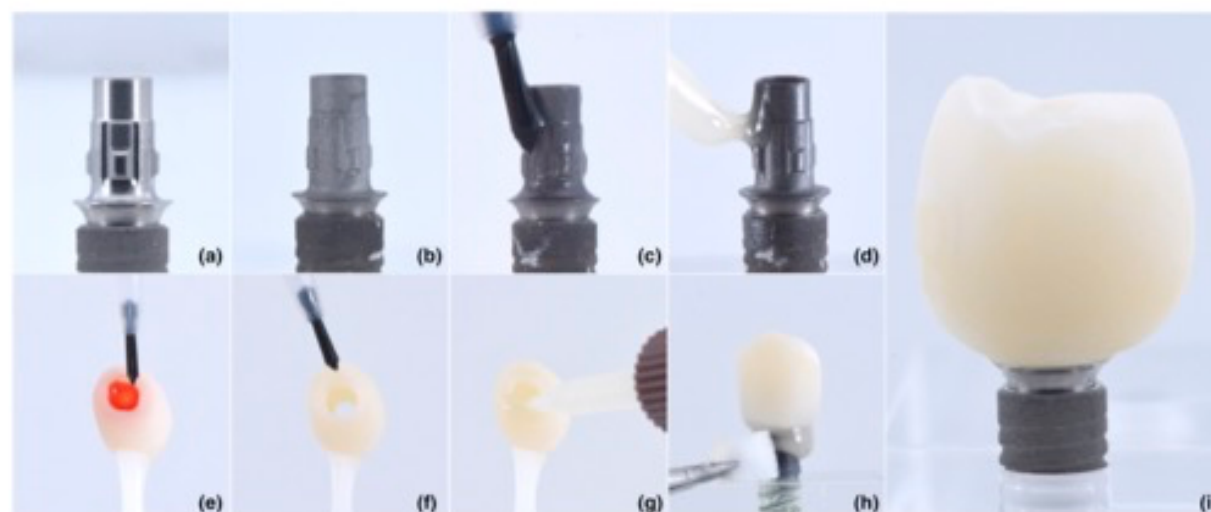


FIGURE 2 Adhesive protocol that was followed for bonding of PICN iSCs to ti-base abutments. (a) Representative specimen from St group; (b) 50- μ m Al_2O_3 sand-blasting under 2.5 bars 20s/cm² and ultrasonic cleaning (4 min with alcohol); (c) primer application (20s) and air dry; (d) resin cement application; (e) Acid etching (5% HCl acid, 60s) and ultrasonic cleaning (4 min with alcohol); (f) primer application (20s application/air dry); (g) Resin cement application; (h) excess removal; (i) representative specimen from St group after cementation of PICN ISC.

The screw access holes were closed with polytetrafluoroethylene (PTFE) tape (Debrunner Acifer AG) and a resin based composite filling material (Tetric EvoCeram, Ivoclar Vivadent).

2.4 | Thermo-mechanical aging

A simultaneous thermocycling (5°C–55°C, dwelling time 120s) and chewing simulation (1,200,000 cycles, 98N 1.67 Hz, CS-4.8, SD Mechatronik) was applied to all specimens. The specimens were loaded from the occlusal surface 2 mm lingual from the screw access channel parallel to the implant axis, and with a vertical movement of 2 mm. Ceramic balls with $\varnothing$ 6 mm (Steatite ball, Hoechst CeramTec) that approximates the Vickers hardness of enamel were fixed to the indenters to be used as antagonists.

2.5 | Complications and outcome measures

After the thermo-mechanical aging, all surviving specimens underwent thorough examination to identify and classify non-catastrophic events such as cracks, debonding, or screw loosening. Two calibrated observers (DK, JP) evaluated these events using a light microscope with 50 $\times$ magnification (Olympus SZX9, Olympus). The outcome measures to identify non-catastrophic events were as following:

- Debonding incidents: micro-movement: when slight movement between the abutment and the crowns is visible under a microscope and loss of retention; macro-movement: the movement is also visible to the naked eye; loss of retention: the complete displacement of the crown from the abutment.
- Screw loosening: any movement between the abutment and the implant.

2.6 | Fracture load test

The survived specimens were statically loaded parallel to the implant axis until failure using a universal testing machine (Shimadzu AGS-X series, Shimadzu). A 0.5 mm thick tin foil (Dentaurum GmbH) was placed between the specimens and the indenter, to ensure even force distribution to the crown/abutment during loading. Failure was defined either as visible fracture of the abutment/ crown, or as 20% decrease of the maximum load during the fracture load test if no fracture was detected. The fracture load of each group was recorded using specific software (TRAPEZIUM X, V.1.4.4, Shimadzu).

2.7 | Failure analysis

During the fracture load test video recordings with magnification (Nikon D250 and Macro Lens) were done, the recordings were

later analyzed to detect if any elastic and/or plastic deformation had happened during the test. Elastic deformation was defined as the reversible distortion of implant-abutment assembly in response to applied force, where the material returned to its original shape once the force is removed whereas plastic deformation was defined as the irreversible alteration of the components resulting in permanent changes. The specimens then were inspected (20 $\times$; 50 $\times$; VHX-6000, Keyence) for determining the mode of failure. Failures, if any, including fractures, visible crack lines, or plastic deformation at implant, abutment, and/or crown level were documented.

An additional scanning electron microscope (SEM) analysis on the selected specimens from each group were done to identify the failure characteristics using an SEM (11 $\times$, 25 $\times$, 50 $\times$, and 200 $\times$, Gemini Zeiss Sigma 300 VP; Carl Zeiss) (Figure 4a–f).

2.8 | Statistical analysis

Shapiro-Wilk tests were used to verify the normal distribution of the mean fracture load values across ti-base abutment types. All study groups showed normal distribution; therefore, a parametric test, one way ANOVA, was performed and followed by Tukey post hoc test to detect the difference across the study groups. The elastic deformations occurrence during load to fracture test was compared by Pearson's chi-square tests. The level of significance was set at $\alpha = .05$ for all the tests. The statistical analysis was performed using a statistical software program (IBM SPSS Statistics, v24.0; IBM Corp).

3 | RESULTS

All specimens survived the thermomechanical aging resulting in 100% survival rate for all groups after 1,200,000 cycles. No visible cracks were observed in any of the implant components or crowns, nor were crown debonding and screw loosening detected. The evaluation of the specimens under magnification presented no micro-movement due to debonding.

The descriptive statistics for the fracture strength values are presented in Table 2. One-way ANOVA test detected a significant difference between the groups ($F(5, 66) = 6.06$ $p < .001$), therefore followed by Tukey post-hoc test. CC-P group showed significantly better fracture strength than CC ($p < .05$) and St ($p < .05$). Furthermore, Th group showed highest strength values among the groups with significant differences compared to Nb-T ($p < .05$), CC ($p < .001$), and St ($p < .001$) groups (Table 2 and Figure 5).

A chi-squared test was applied to find any difference across the groups regarding deformation under static loading (Table 2). The test indicated a difference was found across the groups ($\chi^2 = 23.18$, $p < .001$) (Table 2). The deformation was prominent for CC-P and Nb-V groups, which were accompanied by conical connection ti-base abutments with biggest height and biggest circumferential dimensions, respectively. The deformation observed was of a

plastic nature, manifesting as a bending of the implant-ISC assembly towards the lingual, where the static axial loading was applied. No elastic or plastic deformation under static loading was detected among Th group specimens, which was the only study group with butt-joint implant-abutment connection. The fracture started from the cervical area of the PICN crowns and tracted towards occlusal plane by creating a clear fracture line in majority of the specimens (Figure 3a-f). The fractographic analysis conducted using SEM revealed that there were no notable distinctions in the fracture patterns observed among various ti-base designs (Figure 4a-f).

4 | DISCUSSION

The results of the present study showed that prosthetic height of the ti-base significantly influenced the fracture strength of the ISCs. However, other abutment features, such as pre-packaging sand-blasting of the titanium base surface or the presence of anti-rotational macro-geometrical features, did not demonstrate a significant effect. Importantly, no complications related to ti-base, ISC, or implant were observed during the thermomechanical aging test. Therefore, the null hypothesis was partially rejected.

All specimens survived the thermomechanical aging process without any visible cracks, crown debonding, screw loosening, or debonding-related micro-movement, demonstrating a 100% survival rate. The specimens were loaded with 98 N, which exceeds the expected maximum chewing force of 70 N occurring in the molar area in the absence of parafunctional activity (Bates

et al., 1976; De Boever et al., 1978; Kohyama et al., 2004; Schindler et al., 1998).

Similarly, in a study by Yazigi et al. (2020), which investigated various ISC materials, including zirconia ceramic, lithium disilicate, nano-hybrid composite resin, and PICN, under similar thermocycling conditions and a 49 N loading force, no debonding or mechanical failures were observed during thermomechanical loading. However, it is noteworthy that the same study reported median fracture resistance values of 670 N (IQR: 111) for PICN ISCs, which were significantly lower than the findings of the present study and previous data published on PICN fracture resistance (Yazigi et al., 2020). The test settings in Yazigi et al. (2020)'s study were similar to the present study, as both investigations included thermomechanical aging prior to the fracture resistance test, and the loading was applied parallel to the axial axis of the implant (Yazigi et al., 2020). However, one notable difference was the utilized an occlusal load which was set to 49 N by Yazigi et al., whereas present study used a higher occlusal load of 100 N. Nonetheless, it is important to mention that Yazigi et al. did not provide information about the mode of failure or the fractographic analysis of the specimens, which could have further elucidated the reasons behind the observed differences in fracture resistance values (Yazigi et al., 2020).

On the other hand, Sieper et al. investigated the fracture behavior of PICN and lithium disilicate molar crowns, reporting significantly lower mean fracture strength values for PICN compared to lithium disilicate (Sieper et al., 2017). However, the reported mean fracture strength values for PICN crowns after artificial aging were 2128 N (1 mm thickness) and 2281 N (1.5 mm thickness), both of

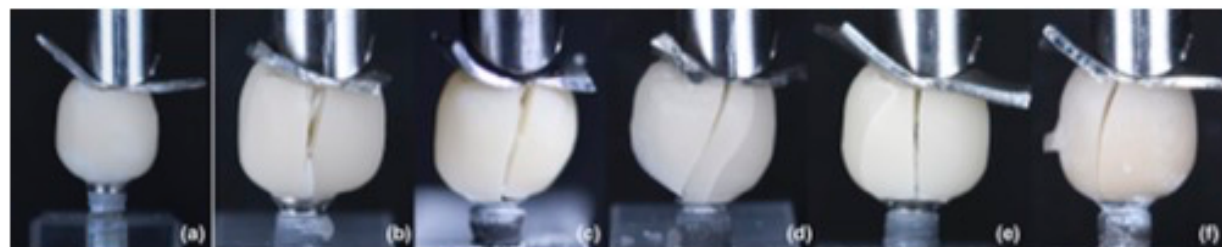


FIGURE 3 Static loading in a universal testing machine and representative images from study groups (a) Th (b) Nb-T; (c) CC; (d) CC-P; (e) Nb-V; (f) St.

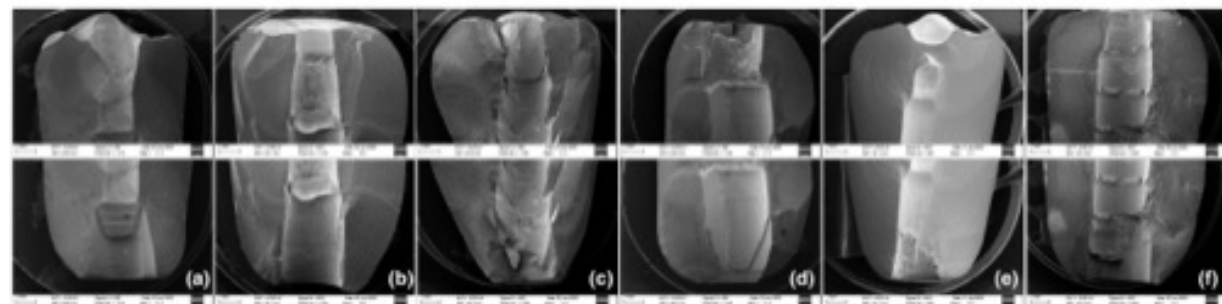


FIGURE 4 SEM images from the representative specimens from study groups. (a) Nb-T; (b) CC; (c) CC-P; (d) Nb-V; (e) St (f) Th.

Fracture strength						Turkey test ^a ($\alpha = .05$)	Deformation ratio
Groups	N	Mean (N)	SD	Median (N)	SE		
Nb	12.00	2231.43	582.90	2097.86	168.27	a, b	0.58
CC	12.00	2029.97	278.87	2026.17	80.50	a	0.75
CC-P	12.00	2554.78	408.03	2442.02	117.79	b, c	0.49
Nb-V	12.00	2316.93	348.90	2362.84	100.72	a, b, c	0.83
St	12.00	2013.92	329.23	1994.49	95.04	a	0.33
Th	12.00	2764.43	477.03	2691.68	137.71	c	0.00

Abbreviations: CC, Camlog Conelog; CC-P, Camlog Conelog prototype; N, Newton; Nb-T, Nobel Biocare Universal Base; Nb-V, Ivoclar Viteobase; St, Straumann Variobase; SD, standard deviation; SE, standard error; Th, Thommen Varioflex.

^aDifferent small letters identifies the significant difference ($\alpha = .05$).

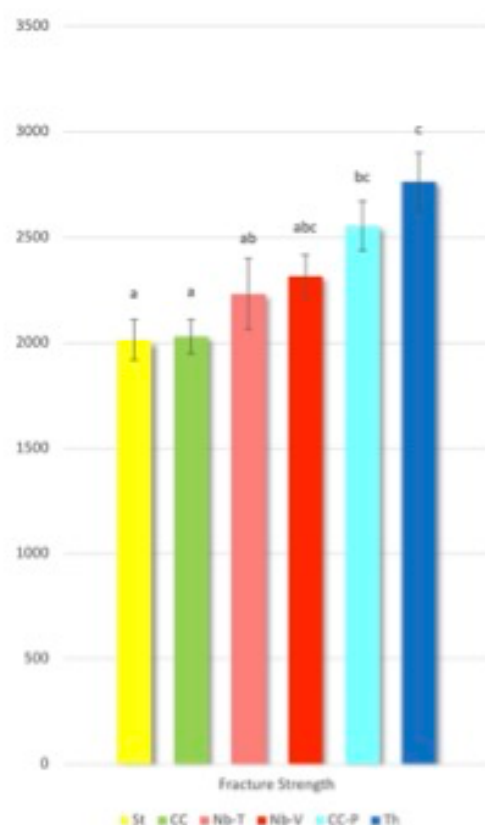


FIGURE 5 Fracture resistance values and Tukey post-hoc test results (Different letters indicates significance difference between groups $\alpha = .05$).

which exceed physiological masticatory forces (Sieper et al., 2017). These results are consistent with the present study.

Undoubtedly, the high success rate observed after the thermocycling test in the present study can be attributed to the bonding protocol followed. Previous research has consistently emphasized the significance of surface treatments (Pitta, Burkhardt, et al., 2021)

and bonding protocols (Burkhardt et al., 2022) in enhancing the mechanical stability of the ISC/ti-base assembly. In the current study, the adhesive protocol adopted from Burkhardt et al., which yielded the best results for bonding PICN ISCs to ti-base abutments. Burkhardt et al. (2022) investigated the bonding of three different restorative materials (PICN, Lithium Disilicate, and zirconia) to ti-bases using three different resin cements (Multilink Hybrid abutment, Panavia V5, and RelyX Ultimate), along with their corresponding primers. Notably, when PICN ISCs were combined the MDP monomer-containing cement system selected in the present study (Clearfil Ceramic Primer Plus and Panavia V5, Kuraray), they exhibited significantly less debonding compared to the other two restorative materials (Burkhardt et al., 2022). The adoption of this specific bonding protocol has contributed to the superior performance of the PICN ISCs in our study, further solidifying the importance of using appropriate resin cements tailored to specific restorative materials for achieving optimal bonding and mechanical stability.

The current study's results also suggest that ti-base prosthetic height is a significant factor in the fracture resistance of the crown-titanium base assembly. Longer abutments (CC-P: prosthetic height (PH) = 7; Th: PH = 9) had a significantly higher fracture strength force than shorter abutments (CC: PH = 4.3 mm and Nb-T: PH = 4.3). Rödiger et al. (2018) investigated the influence of ti-base length as well as the adhesive cement used on the pull-off forces of single implant crowns on ti-bases (Rödiger et al., 2018). In parallel to the present study's findings, increased height of the abutment resulted in a significant increase in the removal forces (Rödiger et al., 2018).

Interestingly, the height of the titanium base had a greater impact on fracture resistance compared to other properties examined, such as pre-packaging sand-blasting of the titanium base surface (Nb-V, Figure 1) or the presence of antirotational macro-geometrical features (such as in Nb-T and St groups, Figure 1) (Table 2). Moreover, neither the anti-rotational features nor the prosthetic height (PH) of the abutment demonstrated influence on mechanical stability, as all specimens withstood the 1,200,000 cycle of thermomechanical loading.

The fracture loads observed in each group of specimens exceeded 2000 N, significantly surpassing the typical masticatory force range of 346–548 N (Karasan et al., 2022). This high strength

can be attributed, in part, to the mechanical properties of the PICN material. Research suggests that PICN materials exhibit improved fracture toughness and machinability compared to ceramics, due to their reduced brittleness, rigidity, and hardness, coupled with enhanced flexibility (Coldea et al., 2013).

Moreover, the application of a hybrid ceramic material with moderately elastic characteristics in iSCs showed a significant capacity to absorb or diminish around 70% of the masticatory forces transmitted to the simulated peri-implant bone (Menini et al., 2013). This notable improvement stands in contrast to the stiffness typically associated with materials, such as zirconia, glass ceramic, and gold. These findings also can explain the 100% survival rate and absence of complications observed during the thermomechanical aging test.

Based on the observed deformations under static loading, it can be suggested that the implant-abutment connection design may influence the distribution of stresses. Notably, the butt-joint implant abutment-connection displayed no deformation under load, suggesting its potential to distribute stress more effectively and minimize its transfer to the implant. In contrast, abutments with larger circumferential dimensions and robust structures were associated with plastic deformation and bending of the implant, which might be explained by greater stress transfer to the implant. While the observed deformations offer valuable insights, it is important to recognize that other parameters may also impact these results. Therefore, drawing conclusions about the causality of deformation patterns and the influence of implant-abutment connections requires careful consideration.

It is important to emphasize that this study examined PICN iSCs, and its findings may not directly apply to other prosthetic materials. The mechanical behavior under stress, bonding strength, the influence of chosen bonding protocol, and machinability can differ among various prosthesis materials. These disparities could potentially affect the mechanical stability of the ti-base/iSC assembly (Patzelt et al., 2015). This is why, standardizing the iSC material was essential to ensure the standardization among the study groups in order to investigate the influence of ti-base geometry. Moreover, the study was conducted in vitro, and the findings should be interpreted with caution.

5 | CONCLUSIONS

Within the scope and limitations of this study, the selection of the ti-base abutment demonstrated a significant influence on the fracture resistance of screw-retained PICN iSCs. Notably, the height of the titanium base was the predominant factor impacting fracture resistance, superseding the influence of other geometric features, namely, anti-rotational features and pre-packaging sandblasting. Interestingly, ti-base abutment characteristics showed no discernible effect on mechanical stability during the aging, as all specimens withstood the thermomechanical loading without complication or failure. Further research is necessary to provide recommendations for optimal ti-base geometry that enhances mechanical stability and prolongs the longevity of implant-supported restorations.

AUTHOR CONTRIBUTIONS

Duygu Karasan: Concept/Design, Methodology, Investigation, Data collection, Data Interpretation, Drafting article, Critical revision of the article. Joao Pitta: Concept/Design, Methodology, Data Interpretation, Critical revision of the article. Cristina Zarauz: Data Interpretation, Critical revision of the article. Malin Stradling: Data Interpretation, Critical revision of the article. Xinran Liu: Investigation, Data collection, Critical revision of the article. Vincent Fehmer, MDT: Concept/Design, Methodology, Investigation, Critical revision of the article. Irena Sailer: Concept/Design, Methodology, Data interpretation, Critical revision of the article, Approval of the submitted and final versions.

ACKNOWLEDGEMENTS

The authors would like to thank to Prof Dr Susan Scherrer from University of Geneva, University Clinics of Dental Medicine, Division of Fixed Prosthodontics and Biomaterials, for her support in SEM analysis and for providing SEM images. An acknowledgment is also given to the manufacturers, such as Biohorizons Camlog, Straumann, Nobel Biocare (Grant no: 2019-1615), Thommen Medical, Ivoclar Vivadent, and Vita Zahnfabrik, for their support with the respective materials.

CONFLICT OF INTEREST STATEMENT

Authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data sets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

ETHICS STATEMENT

Ethical statement is not applicable to this study.

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How to cite this article: Karasan, D., Pitta, J., Zarauz, C., Strasding, M., Liu, X., Fehmer, V., & Sailer, I. (2023). The influence of titanium-base abutment geometry and height on mechanical stability of implant-supported single crowns. *Clinical Oral Implants Research*, 00, 1–9. <https://doi.org/10.1111/clr.14207>